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Program: BTech CSE

HDOC Day-8

Question 1

Write a C program to find the remaining volume of a cuboid after removing a cube of given side.

Step 1: Problem Understanding

Inputs: Length of cuboid (l), Breadth of cuboid (b), Height of cuboid (h), Side of cube (s)

Output: Remaining volume

Formula: Volume of cuboid = $l \times b \times h$, Volume of cube = s^3 ,
Remaining volume = $(l \times b \times h) - s^3$

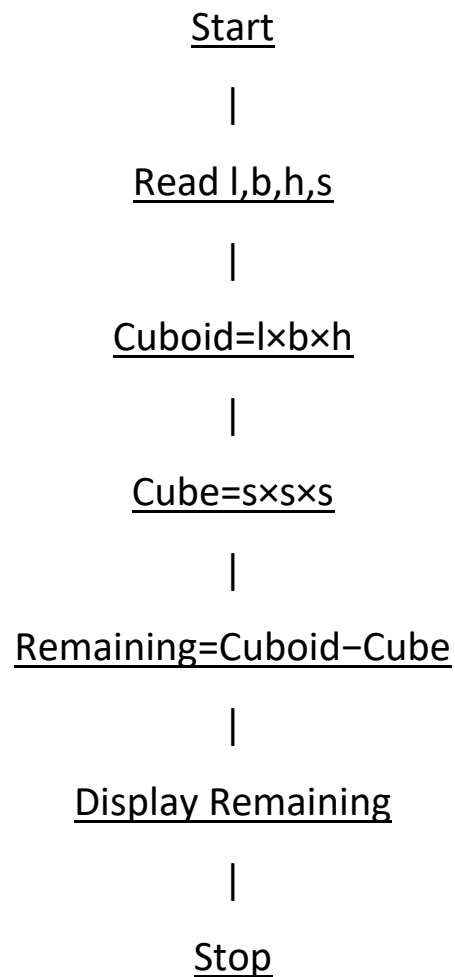
Step 2: Standard Test Cases

Test case	Inputs(l,b,h,s)	Expected output	Actual output	Result
1.	10,5,4,2	192	192	Pass
2.	8,6,5,3	213	213	Pass
3.	12,8,6,4	512	512	Pass

Step 3: Algorithm

1. Start.
2. Read length, breadth, height, and side of cube.
3. Calculate the volume of the cuboid.
4. Calculate the volume of the cube.
5. Subtract cube volume from cuboid volume.
6. Display the remaining volume.
7. Stop.

Step 4: Flowchart



Step 5: C Program

Make “ volume. c” file and then write this code-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float l,b,h,s,cuboid,cube,remaining;
```

```
    printf("Enter length breadth and height of cuboid: ");
```

```
    scanf("%f%f%f",&l,&b,&h);
```

```
    printf("Enter side of cube: ");
```

```
    scanf("%f",&s);
```

```
    cuboid=l*b*h;
```

```
    cube=s*s*s;
```

```
    remaining=cuboid-cube;
```

```
    printf("Remaining volume = %.2f",remaining);
```

```
    return 0;
}
```

Step 6: Code Evaluation

Test Case 1: The program correctly calculated the remaining volume after subtracting the cube's volume from the cuboid's volume. Result: Pass

Test Case 2: The program accepted different dimensions and produced the expected remaining volume accurately. Result: Pass

Test Case 3: The program handled another valid input set correctly and displayed the expected output. Result: Pass

Step 7 : compilation and execution

```
(kali㉿kali)-[~/Desktop]
$ cc volume.c

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter length breadth and height of cuboid: 10 5 4
Enter side of cube: 2
Remaining volume = 192.00

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter length breadth and height of cuboid: 8 6 5
Enter side of cube: 3
Remaining volume = 213.00

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter length breadth and height of cuboid: 12 8 6
Enter side of cube: 4
Remaining volume = 512.00
```

Question 2

Write a C program to find the area of a parallelogram using two adjacent sides and the included angle.

Step 1: Problem Understanding

Inputs: First side (a), Second side (b) , Included angle (θ in degrees)

Output: Area of parallelogram

Formula: $\text{Area} = a \times b \times \sin(\theta)$

Since C's $\sin()$ function uses radians:

$\theta \text{ (rad)} = \theta \times 3.14159/180$

Step 2: Standard Test Cases

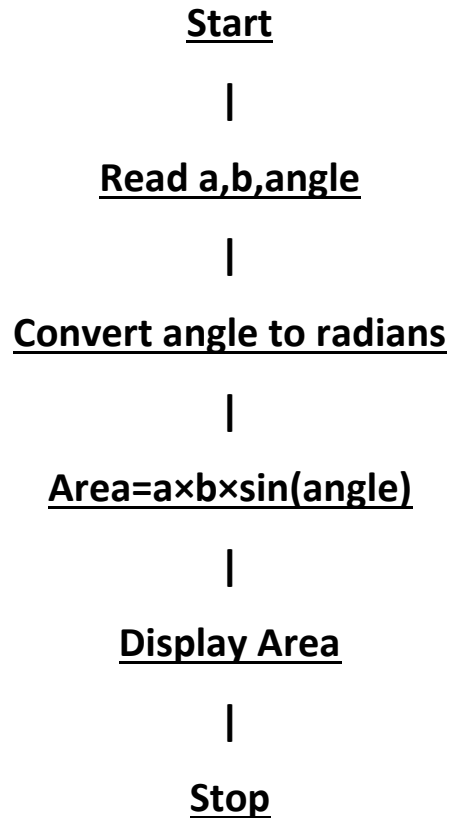
Test case	Inputs(a,b, θ)	Expected output	Actual output	Result
1.	10,5,30(degree)	25.00	25.00	Pass
2.	8,6,90(degree)	48.00	48.00	Pass
3.	12,10,45(degree)	84.85	84.85	Pass

Step 3: Algorithm

1. Start.
2. Read the two sides and included angle.
3. Convert the angle into radians.
4. Calculate the area.

5. Display the area.
 6. Stop.
-

Step 4: Flowchart



Step 5: C Program

Make “ area. c” file and then write this code-

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main()
{
    float a,b,angle,radian,area;

    printf("Enter first side: ");
    scanf("%f",&a);

    printf("Enter second side: ");
    scanf("%f",&b);

    printf("Enter included angle in degrees: ");
    scanf("%f",&angle);

    radian=angle*3.14159/180;

    area=a*b*sin(radian);

    printf("Area of parallelogram = %.2f",area);
    return 0;
}
```

Step 6: Code Evaluation

Evaluation of the Code Against the Test Cases Prepared in Step 2:

Test Case 1: The program correctly calculated the area of the parallelogram using the given sides and included angle. Result: Pass

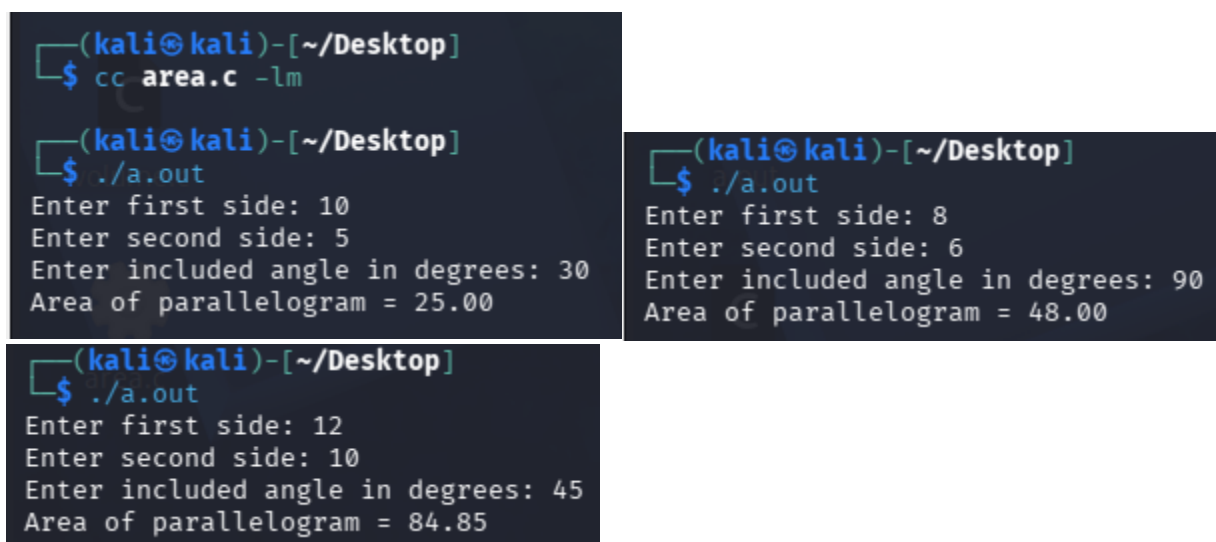
Test Case 2: The program produced the expected area for another set of valid inputs. Result: Pass

Test Case 3: The program correctly handled different input values and displayed the correct area. Result: Pass

Step 7: compilation and execution

Compilation command – “cc area.c -lm”

(-lm is required because the program uses the math.h library.)



```
(kali㉿kali)-[~/Desktop]
$ cc area.c -lm

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter first side: 10
Enter second side: 5
Enter included angle in degrees: 30
Area of parallelogram = 25.00

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter first side: 8
Enter second side: 6
Enter included angle in degrees: 90
Area of parallelogram = 48.00

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter first side: 12
Enter second side: 10
Enter included angle in degrees: 45
Area of parallelogram = 84.85
```